SafeCase: Rapid-Response Kit for Online Harassment

Working name. Hackathon project plan for Problem Statement 1 (Women's Safety, Physical and Digital).

1. The problem in simple words

When a woman is threatened or blackmailed online, or when a morphed image of her is posted, she usually does not know what to do first. She is scared, evidence gets deleted, complaints get rejected for lack of proof, and she often tells no one out of shame.

The main issues women face:

- **Online threats:** abusive messages, blackmail and sextortion, cyberstalking, doxxing
- **Privacy breaches:** leaked private images, stalkerware, location tracking
- **Identity misuse:** fake profiles, morphed and deepfake images, impersonation
- **Physical safety:** street harassment, unsafe travel, no discreet way to call for help
- **The help-seeking gap:** not knowing what evidence to keep, which authority handles what, and complaints that stall after filing

Our project focuses on the last two digital items and the help-seeking gap.

2. Who it is for

A woman aged 18 to 35 who is being threatened or blackmailed on social media or messaging apps, who is stressed, short on time, and unlikely to spend weeks collecting evidence. A friend, sister or counsellor may help her.

3. What already exists, and the gap

Existing solution	What it does	Gap
112 India app, Himmat Plus, Raksha, Safetipin, bSafe	SOS button, location sharing, contact alerts	Focus on the moment of danger, nothing for online harassment or what happens afterwards

Existing solution	What it does	Gap
StopNCII.org	Creates a fingerprint of an intimate image on the user's device so participating platforms can block copies	Intimate images only, not a full case or complaint tool
National Cybercrime Reporting Portal	Official complaint filing	A filing channel, not preparation. Victims report problems with routing and follow-up
Platform report tools	Each platform removes content by its own policy	Separate processes, no single view

The gap we fill: nothing helps a woman act within minutes, preserve proof safely, and arrive at the official channels prepared.

4. Our solution in one line

> An emergency kit for online harassment. She shares what happened, and the app protects her, prepares her reports and keeps her records, while she does almost no work.

We do not replace the police or the cybercrime portal. We make sure she arrives there prepared, with her privacy intact.

5. How it works: 4 steps

```
Share  ->  Understand  ->  Act  ->  Track
```

Step 1: Share. She shares the threatening message or the link to a posted image to the app. One tap, nothing to type.

Step 2: Understand. The app tells her in a few words what this is and how serious it is, with reasons. Example: "This looks like blackmail. High risk. Don't pay. Don't delete anything. Call 112 if you feel in danger."

Step 3: Act. The app gives ready-made actions:

- Report to the platform (pre-filled notice)
- Protect the image from spreading (the image stays on her phone)
- Complaint pack for the cybercrime portal (ready to copy across)

Step 4: Track. Every new message she shares is added automatically. The app shows what changed (for example, "now includes a threat") and reminds her to follow up.

6. If the harasser posts an image

We cannot stop him from posting. We shrink the time between "posted" and "reported" from hours of confusion to a couple of minutes:

1. She shares the post link. She never has to look at the image again.
2. The app generates the platform report and the grievance-officer notice, pre-filled.
3. It starts a timer and drafts an escalation if there is no response by the deadline.
4. It adds the link and fingerprint to the case automatically.
5. For intimate images, it guides her through StopNCII for any remaining copies.

Preventive Shield (optional mode): before anything happens, a woman can calmly fingerprint sensitive images, run an account lockdown checklist (private profile, two-factor authentication, tagged photos) and prepare notices in advance.

7. The dashboard: one screen, five parts

```
+---------------------------------+
|  My Safety Case                 |
|                                 |
|  RISK LEVEL:  HIGH              |
|  (blackmail, money demand)      |
|                                 |
|  [ + Add new message / link ]   |
|                                 |
|  WHAT TO DO NOW                 |
|  [ ] Report profile on Insta    |
|  [ ] Block photo (StopNCII)     |
|  [ ] File cybercrime complaint  |
|                                 |
|  TIMELINE                       |
|  Mon  Insults                   |
|  Wed  Repeated messages         |
```

```
|  Fri  Threat + money demand    |
|                                |
|  [ Download complaint pack ]   |
|  [ Need help now? Call 112 ]   |
+--------------------------------+
```

From top to bottom:

1. **Risk level:** one color, one sentence
2. **Add button:** the main thing she will tap
3. **What to do now:** a short checklist
4. **Timeline:** what happened and when
5. **Two buttons:** the complaint document and emergency help

A small **Quick Exit** button sits in the corner and hides the app instantly.

8. What happens behind the screen

Three simple parts:

1. **Reads the message** and pulls out the key facts (threat type, money demand, phone number, handle, deadline, platform)
2. **Stores it safely** with a date, time and fingerprint (hash) so it cannot be edited later without detection
3. **Writes the documents** (platform notice, complaint summary)

The case timeline, the "what changed" view and the complaint pack all come from the same stored case.

9. Keeping her data protected

If she shares a link

- Only the link and date are stored. The app does not open it, download the image or show a preview.
- The link is encrypted on her phone, and only her PIN unlocks it.

If she shares an image

- The image never leaves her phone. The phone creates a fingerprint, and only the fingerprint is used.

- The server never receives the image, so it cannot leak it.
- If she keeps a copy as evidence, it stays encrypted on her own device and is blurred by default.
- No AI is ever run on the image.

For everything stored

- PIN or fingerprint lock
- Encryption of stored data
- One-tap "Delete everything"
- Minimum data: no name, contacts or location unless she chooses to add them
- She decides exactly what goes into a complaint (for example, the link only, without the image)

10. Making her feel safe

1. **Say it clearly at the right moment.** When she shares an image: "This stays on your phone. We never see it. Only you can open this."
2. **Do not show harmful content back to her.** Use a blurred thumbnail with tap-to-reveal and a text summary.
3. **Hide the app.** Quick Exit, an optional neutral app name and icon, and notifications that reveal nothing.
4. **Use calm, kind language.** For example: "You did the right thing. This is not your fault. Here is what we can do together." No alarming screens except the 112 button when it is truly urgent.
5. **Let her control the pace.** She chooses what to add, skip and share. Nothing is forwarded automatically.
6. **Offer a trusted helper.** An optional friend, sister or counsellor who can handle the steps so she does not have to view the content again. Off by default, and she sets what the helper can see.
7. **Be honest about limits.** "We help you act fast and keep your information private. We cannot guarantee removal, but you will not be doing this alone."

A known risk: a safety app can be misused if a controlling person gets access to her phone. Quick Exit, a neutral name, hidden notifications and the PIN are the mitigation. Say this in the pitch.

11. What to build for the hackathon (MVP)

Only these five things:

1. Add a message or link (paste or share)
2. Risk level with reasons
3. Checklist of next steps, with the urgent help card on high risk
4. Timeline that updates when more is added
5. Downloadable complaint document

Supporting pieces, kept small: PIN lock, Quick Exit, hash and timestamp on every item, local fingerprinting of images.

Suggested technology:

- Frontend: mobile-first web app (React or Next.js)
- Backend: FastAPI or Node
- Database: SQLite or Postgres
- AI: Claude API for message classification (text only), with a strict JSON output and guardrails
- Hashing: SHA-256 in the browser, perceptual hash for image matching
- PDF: server-side generation

12. Future work (put on a slide, do not build now)

- Digital Exposure Graph, with an "attacker could combine this" view
- Threat Linkage across cases, limited to explainable matches (same phone number, handle, link or UPI ID), always worded "possibly related"
- Scam-template triage, to tell a mass bluff script from a targeted threat
- Follow-up tracker with complaint numbers and escalation drafts
- Digital Safety Autopsy: explaining how the incident developed and what exposure was preventable
- Physical-safety add-on: check-in timer with contact alerts

13. Honest limits

- We cannot stop a harasser from posting.
- We cannot guarantee platform removal or police action.

- Fingerprint matching only catches known copies on participating platforms, not edited versions.
- We do not claim to detect deepfakes in general.
- Risk ratings are guidance. A human should review them, and the AI never gives a legal conclusion.
- We do not submit complaints automatically. We prepare a pack she can use on the official channels.

14. Demo plan (3 to 4 minutes)

Use clearly labeled simulated data. Never use real people's images or messages.

1. **Persona (30 sec):** "Meet Ananya, 22. Someone is blackmailing her."
2. **Share:** she shares one threatening message. Show the risk level, the reasons and the urgent help card.
3. **Privacy moment:** she selects an image, and the banner reads "This stays on your device." Show the fingerprint, with the blurred thumbnail.
4. **Simulate a post:** the harasser's post link arrives. The app produces the takedown notice in one tap, and the timeline updates itself.
5. **Complaint pack:** open the downloadable document.
6. **Close:** Quick Exit, "Delete everything," and the roadmap slide.

Keep a backup screen recording in case of network problems.

15. How this maps to the judging rubric

Criterion	Weight	Our answer
Problem Relevance and Understanding	15%	One persona, one clear gap around the official channels, researched existing solutions
Innovation and Originality	15%	Rapid response first, evidence that builds itself, privacy-first handling of sensitive images

Criterion	Weight	Our answer
Technical Implementation	20%	Signal extraction, hash chain, on-device hashing, complaint generation, tests for hashing and output schema
Working MVP and Demonstration	25%	One unbroken flow: share, understand, act, track
Feasibility and Potential Impact	15%	Partners such as NGOs, helplines, legal-aid clinics and college grievance cells, plus a validation plan
User Experience and Accessibility	10%	Few taps, calm language, Quick Exit, multilingual and low-literacy friendly

16. The pitch in one line

> "When a woman is threatened online, we help her act in minutes. The image never leaves her phone, only she holds the key, and the evidence collects itself."

17. To verify before presenting

These came from general knowledge or older articles. Check them on official sources before they appear on a slide or inside the app:

- Helpline numbers (112, 181, 1930) and the current cybercrime portal process
- Platform takedown deadlines under the current IT rules, and each platform's grievance-officer contact
- Legal sections that apply (IT Act, Bharatiya Nyaya Sanhita), ideally reviewed by a lawyer or legal-aid clinic
- Whether StopNCII covers morphed or AI-generated images
- Current features of the apps in the comparison table (install two or three yourself)